Scale of the Universe

I. A Sense of Scale

"The universe is bigger than you think. No—it's bigger than that, too. The problem with scale is that your brain wasn't built for it. You evolved to understand the distance between a tree and a lion, not the distance between here and the edge of the observable universe. So tonight, we're going to try anyway."

The name tags are kept in a beige plastic box in the observatory office, sorted alphabetically. I flip to H and find mine—*Lyra Huang, UVA Astronomy*—printed in neat serif beneath the department seal, a cartoon observatory against a field of stars. I clip it to my jacket and walk inside.

Three years ago, the box held no tag with my name. I was a first-year who had signed up as a volunteer for Public Night on a whim, arriving twenty minutes early in a coat too thin for the walk up Observatory Hill. The volunteers' tags were already laid out, names I didn't recognize, and I stood there scanning them until I found one that read *Huang*—a different Huang, someone I'd never met—and pinned it to my chest. Close enough. I spent the evening answering visitors' questions about things I'd only half-learned, wearing another student's name like a borrowed constellation.

Now it is the last Friday of March, and I am back. I have volunteered here once or twice since then. The lecture hall smells the same—old wood and cold air. Professor Whittle stands at the front, clicking to his first slide. The outline hasn't changed: *A Sense of Scale. Witnessing History. The Evolution of Galaxies. Seeing the Fireball Era. The Creation Mechanism.* I've heard this talk before. I could probably give it myself. But I sit in the back row and listen anyway.

II. Witnessing History

"When you look at something far away, you are looking back in time. The light from the sun takes eight minutes to reach you. The light from the nearest star takes four years. So when you look at the night sky, you are not seeing the universe as it is. You are seeing it as it was. Every telescope is a time machine."

The first time I heard this lecture, I wasn't sitting inside. It was the fall of my first year, and Professor Whittle was also the one at the podium. I stood outside the back door of the lecture hall with a graduate student I'd just met, the two of us leaning against the doorframe while his voice carried through the wall, muffled into something soft and rhythmic, like weather.

She was tall with her hair red and she studied supernovae—dying stars, she said, or maybe not dying, maybe just becoming something else very quickly. I asked her too many questions. She answered all of them. At some point we got to talking about where we were from, and when she found out I was Chinese she told me she'd studied some Mandarin. She'd read *Romance of the Three Kingdoms*. Her favorite character was Cao Cao.

I remember thinking that this was not the kind of conversation I'd expected to have at an observatory in Virginia, leaning against a doorframe on a Friday night, the lecture on the other side of the wall already moving on to the evolution of galaxies—

"So how did you end up in astronomy?"

The girl asking me this is a first-year. She's standing by the telescope with her hands in her jacket pockets and no tag clipped to her collar. I tell her a long story she probably won't fully remember. She

tells me she's thinking about majoring in astronomy. We go through the rest of it—what year are you, where are you from, do you like UVA, why did you come here—the kind of questions that only get asked between people who have just met and are trying to find the shape of each other.

She asks me if the observatory is always this cold, and I say yes, it's the walls—the building is old, the brick holds the chill. She asks me if I've ever seen anything really good through the telescope, and I tell her about the time I saw Saturn's rings so clearly that I thought someone had taped a picture to the lens. She laughs. She asks me what it's like being an astronomy major, whether the classes are hard, whether the professors are nice. I answer all of them.

At some point I realize I am standing exactly where I stood three years ago, on the same side of the same hallway, except that I am on the other end of the conversation now. The wood floor creaks in the same place it always has.

III. The Evolution of Galaxies

"Galaxies don't stay the same. They collide, they merge, they eat each other. The Milky Way itself is a product of dozens of smaller galaxies that fell into it over billions of years. Everything you see in the sky is in the process of becoming something else. It just happens too slowly for you to notice."

The first-year girl asks me where the bathroom is. I tell her it's in the basement, and that it's a little hard to find, and that the basement is kind of unsettling, so I'll walk her down.

The stairs are narrow and the light at the bottom has always been the same reluctant yellow. When I was here as a first-year the basement was full of dust and roped-off rooms and, supposedly, mice. It still smells like dust, though someone has swept. I point her to the bathroom and wait in the corridor.

The last time I stood in this corridor I was with the AstroFun Club, and someone had a key to the archive room. Inside: filing cabinets. An entire room of filing cabinets. They held the parallax records—decades of distance measurements, star by star, logged and filed by hand. I opened one drawer and found stacks of letters between astronomers, written in a cursive so ornate I couldn't read a single sentence. I felt a little guilty, going through someone else's correspondence. But they were all dead, and the letters were beautiful even illegible, and no one stopped me, so I kept opening drawers.

In the back of the room, leaning against the wall behind a cabinet, I found a rubbing of a Chinese star chart. The label said *Tianwen Tu*—a copy of a Song dynasty planisphere carved in Suzhou in 1247, eight hundred years ago, a map of the sky made for the education of an emperor's son. It recorded 1,434 stars. The text at the bottom was in classical Chinese. I could read it. I stood there for a long time.

I took a photo of it that night and spent the next few weeks translating the inscription. This was harder than it sounds. The text had no punctuation—just columns of traditional characters running top to bottom, carved eight centuries ago for someone who would have known where one sentence ended and the next began. I didn't. So I copied it out by hand, character by character, and added the punctuation myself. Then I converted the traditional characters to simplified. Then I translated the classical Chinese into modern Chinese. Then I translated the modern Chinese into English. Four steps to cross a distance that was not measured in light-years but might as well have been.

We come back up the stairs and turn right into the museum room. The girl is saying something about the mice, and I am half-listening because I have already seen it—the *Tianwen Tu*, framed now, hung on the white wall under a proper light. It looks different here. In the basement it had been leaning against a cabinet, gathering dust behind someone's old filing; now it is mounted in a black frame, the

constellations clean and visible, the classical characters running down like a heavy rain that hasn't hit the ground yet.

There is a small plaque beside it. They didn't put the full translation up—the inscription runs over two thousand characters, so of course they didn't—but they used the introduction I wrote to go with it. The girl reads the plaque. She doesn't say anything about it. There's no reason she would.

IV. Seeing the Fireball Era

"If you look far enough, you can see the beginning. Not a metaphor—I mean literally. The cosmic microwave background is the oldest light in the universe. It has been traveling for thirteen point eight billion years. When it started, there were no stars, no galaxies, no planets. Just light, everywhere, in every direction. A fireball the size of everything."

After the lecture, Professor Murphy takes the visitors from Professor Whittle and leads them to the dome room. I've seen this handoff happen many times. It is always the same—Whittle finishes his last slide, Murphy opens the door to the dome, and the visitors file through with their necks already craning upward.

The telescope is enormous. It doesn't matter how many times I see it; it is always enormous. Murphy tells them what he always tells them—that the dome rotates, that the slit opens to the sky, that the Clark refractor was once the second largest in the world, that it was built in 1885 and has been looking at the same sky ever since. He cranks something and the dome begins to turn, slow and heavy, and the visitors take out their phones.

I have sat in that tall chair before, the one that rises on a rail like a staircase. I've pressed my eye to the eyepiece and held my breath without meaning to. You have to hold still. The telescope is so large and so precise that your own heartbeat will shake the image, and so you learn to make yourself small and quiet, smaller and quieter than you have ever been in a room, because otherwise you'll lose it.

I watched Saturn in the dark. Its rings were fine as a line drawn in pencil, tilted just slightly, as if the planet were showing me its good side. The light I was seeing had left Saturn about an hour and twenty minutes ago—not ancient, not mythological, just an hour and twenty minutes, the length of a movie, the length of a lecture on the scale of the universe. Jupiter's storm was older. The light from the Orion Nebula was older still. And somewhere past the reach of this telescope, past the reach of every telescope ever built, there was light still traveling that had left its source before the Earth existed. I was never going to see it. But it was there, moving toward me, and that was enough.

I have never operated the telescope myself. I've thought about asking. But it is a twenty-six-inch Clark refractor, older than anyone alive, and it costs more than everything I have ever owned, so I haven't.

V. The Creation Mechanism

"So how did all of this get here? That's the last question. The honest answer is: we're still working on it. But here's what we think we know."

Public Night ends the way it always ends. The visitors leave in small groups, talking about what they saw. The first-year girl waves goodbye at the door and drives down the hill toward campus, her

headlights bobbing between the trees. I stay behind to help fold the chairs in the lecture hall and put the name tags back in the box. I file mine under H, between the plastic dividers, and close the lid.

Next year I will graduate. I don't know if I'll come back after that. I'll probably say I will. People always say they'll come back, and sometimes they do, but the building won't know the difference. The dome will keep turning. Professor Murphy will keep giving the lecture. Some other first-year will clip on a volunteer tag and stand by the telescope with their hands in their pockets, asking someone the same questions I once asked, receiving the same answers. The wood floor will creak in the same place it always has, under different shoes.

I think about this sometimes—what stays. My name tag in the box. A conversation that a graduate student has probably forgotten. A conversation that a first-year will probably forget. The floors have been walked on so many times that I imagine everyone who has ever been here has left something behind on them, a kind of impression, the way light leaves an image on the back of your eye after you close it.

I don't have a religion. I'm not sure I'm a materialist either. But I have spent most of my life thinking about where people go when they die. It's the kind of question you can think about for years without getting anywhere, and I have, since I was old enough to understand that the question existed. Somewhere between my first year and now—maybe it was the night I first heard the lecture on the scale of the universe, standing outside the back door with a girl who liked Cao Cao—I arrived at an answer that finally convinced me.

When you die, you go to the edge of the universe.

This is how it works. Light takes time to travel. The light from this moment—me standing in the dome room, the telescope still warm from an evening of use, the chairs half-folded in the lecture hall—this light is already leaving. It will pass the moon in just over a second. It will reach Saturn in about an hour and twenty minutes. In a little over four years, it will pass the nearest star. In two and a half million years it will reach Andromeda. And it will keep going. It doesn't stop. Every moment that has ever happened on Earth is still out there, expanding into the dark, a shell of light growing wider and wider at the speed of light.

So if you could go far enough—impossibly far, further than any telescope has ever looked—you could find the place where the light from your birth has not yet arrived. And if you stood there, and if you had good enough eyes, you could watch yourself being born. You could watch your whole life unfold, scene by scene, at the speed of light. You could watch yourself walk up Observatory Hill for the first time in a coat too thin for October. You could watch yourself pin someone else's name to your chest. You could watch yourself lean against a doorframe, asking a stranger about supernovae.

I don't know if this counts as an afterlife. It's probably not even good physics. But it's the most comforting thing I've ever made up, and I made it up here, in a building that has spent a hundred and forty years collecting other people's light.

I turn off the lights in the lecture hall and walk outside. Professor Murphy locks the dome and offers me a ride home. He drives a Porsche, which I did not expect the first time I saw it and which I still find unreasonably funny. We pull out of the gravel lot and down the hill, the observatory shrinking in the side mirror. He asks me how my classes are going. I tell him fine. Outside, the sky over Observatory Hill is clear and full, and the light from tonight is already leaving.

It has a long way to go. So do I.